Fibonacci Polynomials & *Lights Out*

William Boyles

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Abstract

Lights Out is a game on a simple graph $G = (V, E)$ where clicking a vertex toggles the on/off state of it and its neighbors. Identifying subsets of V with their indicator vectors over $\mathbb{F}_2$, let $\Phi_G : \mathbb{F}_2^V \to \mathbb{F}_2^V$ be the function which maps the set of vertices clicked to the set of vertices that change state. For the $n \times n$ grid, define $d(n) = \dim(\ker \Phi_G)$. We recover the known fact that $d(n)$ is always even and prove infinitely many even values are also impossible. We provide formulas and bounds for $d(n)$, including a complete characterization when $n = 2^a p^b - 1$ for $a, b \geq 0$ and p not a Wieferich prime.

1 Fibonacci Polynomials

We begin by introducing Fibonacci polynomials. We first develop some basic properties of these polynomials that will be useful in relating them to the Lights Out problem.

Definition 1.1 (Fibonacci Polynomial). *Let $F_n(x)$ be the polynomial in the ring $\mathbb{F}_2[x]$ defined recursively by*

$$F_n(x) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ xF_{n-1}(x) + F_{n-2}(x) & \textit{otherwise} \end{cases}.$$

These polynomials are called "Fibonacci" because $F_n(1)$ is the nth Fibonacci number when evaluated over $\mathbb{Z}[x]$. Evaluated over $\mathbb{F}_2$, $F_n(1)$ is the nth Fibonacci number modulo 2.

In $\mathbb{F}_2[x]$, the coefficients of $F_n(x)$ are either 1 or 0, so we can visualize them by coloring squares black or white to represent the coefficients. For example,

$$F_6(x) = 1x^5 + 0x^4 + 0x^3 + 0x^2 + 1x + 0 = \blacksquare\square\square\square\blacksquare\square.$$

Plotting $F_1(x)$ through $F_{128}(x)$ in the same way, aligning terms of the same degree, we see a Sierpinski triangle.

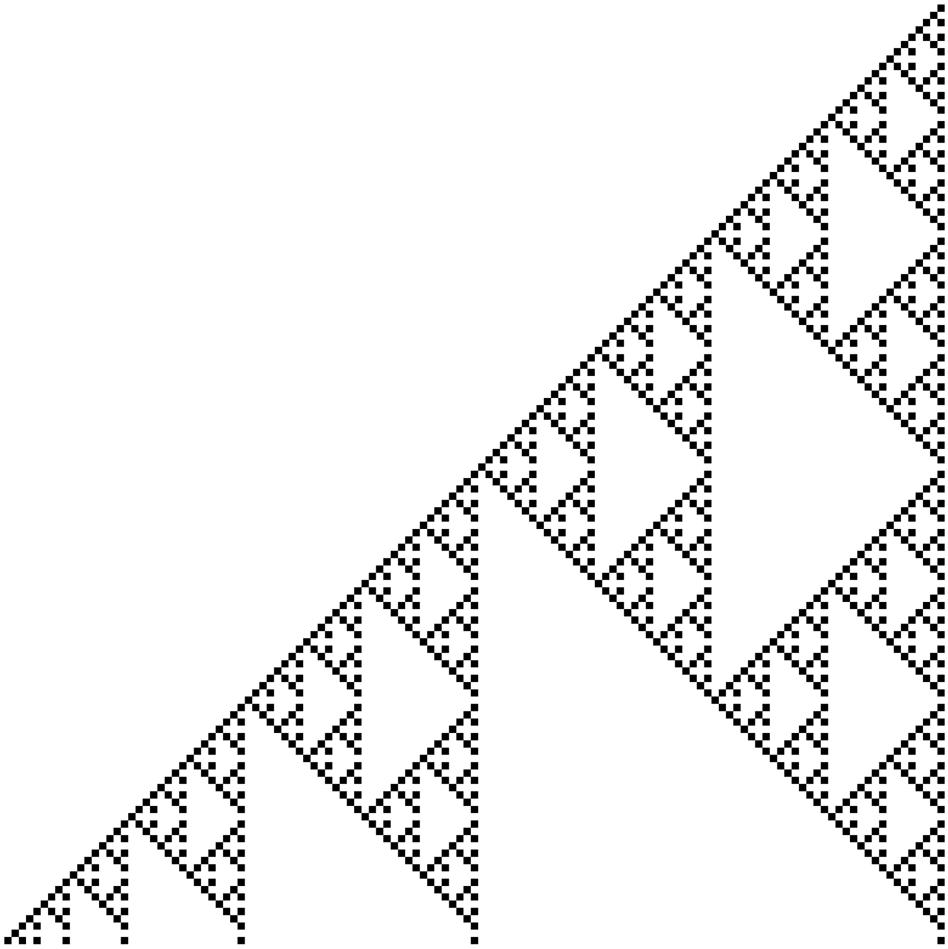

Figure 1: $F_1(x)$ through $F_{128}(x)$, right-aligned.

The Sierpinski arises from the parity of the binomial coefficients in Pascal's triangle, giving the formula

$$F_n(x) = \sum_{i=0}^{n-1} \binom{n+i}{2i+1} x^i \mod 2.$$

Sutner also showed the above construction and formula, along with the following relationship between rectangular grids and Fibonacci Polynomials [Sut00, Theorem 5.1].

Lemma 1.2 (Sutner). *Let $G = (V, E)$ be an $n \times m$ grid graph. Then*

$$\dim(\ker \Phi_G) = \deg\left(\gcd\left(F_{n+1}(x), F_{m+1}(x+1)\right)\right) = \deg\left(\gcd\left(F_{n+1}(x+1), F_{m+1}(x)\right)\right).$$

In particular, this lemma allows us to define $d(n)$ in terms of Fibonacci polynomials.

$$d(n) = \deg\left(\gcd\left(F_{n+1}(x), F_{n+1}(x+1)\right)\right).$$

Sutner and others also established several divisibility properties of Fibonacci polynomials [HMP04, Proposition 2.2][Sut00, Proposition 2.1][Knu11, Section 7.1.3, Exercises 190–194].

Lemma 1.3 (Hunziker, Machiavelo, and Park). *A polynomial $\tau(x)$ in $\mathbb{F}_2[x]$ divides both $F_n(x)$ and $F_m(x)$ if and only if it divides $F_{\gcd(m,n)}(x)$. In particular,*

$$\gcd\left(F_m(x), F_n(x)\right) = F_{\gcd(m,n)}(x).$$

Corollary 1.4. *The following are true:*

(i) x *divides* $F_n(x)$ *if and only if* $n \equiv 0 \mod 2$.

(ii) $x+1$ *divides* $F_n(x+1)$ *if and only if* $n \equiv 0 \mod 2$.

(iii) $x+1$ *divides* $F_n(x)$ *if and only if* $n \equiv 0 \mod 3$.

(iv) x *divides* $F_n(x+1)$ *if and only if* $n \equiv 0 \mod 3$.

Proof. Notice,

(i) We find that $F_2(x) = x$ is the smallest Fibonacci polynomial divisible by x, so we apply Lemma 1.3 to get the desired result.

(ii) Follows from (i) by substituting $x+1$ for x.

(iii) We find that $F_3(x) = (x+1)^2$ is the smallest Fibonacci polynomial divisible by $x+1$, so we apply Lemma 1.3 to get the desired result.

(iv) Follows from (iii) by substituting $x+1$ for x.

■

Hunziker, Machiavelo, and Park give the following characterization of the square-free factors of Fibonacci polynomials [HMP04, Proposition 2.11].

Lemma 1.5 (Hunziker, Machiavelo, and Park)**.** *If* n *is odd, then* $F_n(x)$ *is the square of a square-free polynomial.*

Lemma 1.6. *Let* $P \in \mathbb{F}_2[x]$ *such that* $P(x) = P(x+1)$*. Then* $P \in \mathbb{F}_2[x^2+x]$*.*

Proof. Let $y = x(x+1)$. Every polynomial in $\mathbb{F}_2[x]$ can be written uniquely in the following form

$$P(x) = A(y) + xB(y) \qquad A, B \in \mathbb{F}_2[y].$$

Since P is unchanged by translation,

$$\begin{aligned} P(x+1) &= P(x) \\ A(y) + (x+1)B(y) &= A(y) + xB(y) \\ B(y) &= 0. \end{aligned}$$

Thus, $P(x) = A(y)$ and is thus in $\mathbb{F}_2[y] = \mathbb{F}_2[x^2+x]$, as desired. ■

We can now use these properties to prove several facts about $d(n)$.

Lemma 1.7. *Let* $d(n) = \dim(\ker \Phi_G)$ *for* G *an* $n \times n$ *grid graph. Then* $d(n)$ *is always even.*

Proof. Let

$$G_n(x) = \gcd\left(F_n(x), F_n(x+1)\right).$$

By Lemma 1.2, we know that $d(n) = \deg G_{n+1}(x)$ and $G_n(x) = G_n(x+1)$. Since $G_n(x)$ is unchanged by translation, Lemma 1.6 tells us there exists $H_n \in \mathbb{F}_2[x]$ such that

$$G_n(x) = H_n(x^2+x).$$

So,

$$d(n) = \deg G_{n+1}(x) = 2 \deg H_{n+1}(x)$$

and thus $d(n)$ is even. ■

Sutner observed the above lemma computationally [Sut89].

Corollary 1.8. *If n is even, then $d(n)$ is divisible by 4.*

Proof. If n is even, then $n+1$ is odd. Then by Lemma 1.5, $F_{n+1}(x)$ and $F_{n+1}(x+1)$ are squares of square-free polynomials $\tau(x)$ and $\tau(x+1)$.

$$\begin{aligned} G_{n+1}(x) &= \gcd\left(F_{n+1}(x), F_{n+1}(x+1)\right) \\ &= \gcd\left(\tau_{n+1}(x)^2, \tau_{n+1}(x+1)^2\right) \\ &= \gcd\left(\tau_{n+1}(x), \tau_{n+1}(x+1)\right)^2 \\ \sqrt{G_{n+1}(x)} &= \gcd\left(\tau_{n+1}(x), \tau_{n+1}(x+1)\right) \end{aligned}$$

Since translation by 1 exchanges the arguments of the GCD, $\sqrt{G_{n+1}(x)} = \sqrt{G_{n+1}(x+1)}$. So, by Lemma 1.7, there exists $H_{n+1} \in \mathbb{F}_2[x]$ such that

$$\begin{aligned} \sqrt{G_{n+1}(x)} &= H_{n+1}(x^2+x) \\ G_{n+1}(x) &= H_{n+1}(x^2+x)^2. \end{aligned}$$

So,

$$d(n) = \deg G_{n+1}(x) = 4 \deg H_{n+1}(x),$$

as desired. ■

Note that the implication does not hold in the other direction: $d(n)$ may also be divisible by 4 when n is odd. For example, $d(9) = 8$.

Lemma 1.9. *Let $x = t + t^{-1}$. Then for $t \neq 0$,*

$$xF_n(x) = t^n + t^{-n}.$$

Proof. Define $G_n(t) = t^{n+1} + t^{-(n+1)}$. Then

$$\begin{aligned} xG_{n-1}(t) + G_{n-2}(t) &= (t+t^{-1})(t^n + t^{-n}) + (t^{n-1} + t^{-(n-1)}) \\ &= t^{n+1} + t^{-(n-1)} + t^{n-1} + t^{-(n+1)} + t^{n-1} + t^{-(n-1)} \\ &= t^{n+1} + t^{-(n+1)} \\ &= G_n. \end{aligned}$$

Thus we see that $G_n(t)$ satisfies the same recurrence relation as $F_n(x)$. $xF_n(x)$ also satisfies the same recurrence relation.

$$\begin{aligned} xF_n(x) &= x(xF_{n-1}(x) + F_{n-2}(x)) \\ &= x(xF_{n-1}(x)) + xF_{n-2}(x). \end{aligned}$$

Now we compute $xF_1(x)$ and $xF_2(x)$ in terms of t and G:

$$\begin{aligned} xF_1(x) &= x \\ &= t + t^{-1} \\ &= G_0(t), \end{aligned}$$

and

$$\begin{aligned} xF_2(x) &= x^2 \\ &= (t+t^{-1})^2 \\ &= t^2 + t^{-2} \\ &= G_1(t). \end{aligned}$$

Since the first two terms match and they satisfy the same recurrence relation then we've shown by induction that for $t \neq 0$,

$$xF_n(x) = G_{n-1}(t) = t^n + t^{-n}.$$

■

This was also proven by Hunziker, Machiavelo, and Park [HMP04, Proposition 2.3].

The following identity is a specialization of a result of Hunziker, Machiavelo, and Park [HMP04, Lemma 2.6].

Lemma 1.10 (Hunziker, Machiavelo, and Park)**.** *Let $n = 2^k b$ where $k \geq 0$ and b is odd. Then*

$$F_n(x) = x^{2^k - 1} F_b(x)^{2^k}.$$

Lemma 1.11. *The distinct roots of $F_n(x)$ in $\overline{\mathbb{F}_2}$ are $\{\zeta + \zeta^{-1} : \zeta^n = 1, \zeta \neq 1\}$ together with 0 when n is even.*

Proof. Notice that $F_n(0) = F_{n-2}(0)$, so

$$F_n(0) = \begin{cases} 1 & n \text{ odd} \\ 0 & n \text{ even} \end{cases}.$$

Thus 0 is a root of $F_n(x)$ whenever n is even.

Let α be a non-zero root of F_n. Choose ζ satisfying

$$\zeta^2 + \alpha\zeta + 1 = 0.$$

Notice that ζ cannot be 0. Dividing by ζ,

$$\begin{aligned} \zeta + \alpha + \zeta^{-1} &= 0 \\ \alpha &= \zeta + \zeta^{-1}. \end{aligned}$$

By, Lemma 1.9,

$$\begin{aligned} 0 &= \alpha F_n(\alpha) \\ &= \zeta^n + \zeta^{-n} \end{aligned}$$

Since a field has no nonzero nilpotents, we may multiply by ζ^n.

$$\begin{aligned} 0 &= \zeta^{2n} + 1 \\ &= (\zeta^n + 1)^2, \end{aligned}$$

and

$$\begin{aligned} 0 &= \zeta^n + 1 \\ \zeta^n &= 1. \end{aligned}$$

Thus the order of ζ divides n. Notice too that $\zeta \neq 1$; otherwise, $\alpha = 1 + 1^{-1} = 0$, which we already showed as not possible.

Conversely, if $\zeta^n = 1$ and $\zeta \neq 1$, then let $\alpha = \zeta + \zeta^{-1}$. Then we see $\alpha \neq 0$ and

$$\alpha F_n(\alpha) = \zeta^n + \zeta^{-n} = 1 + 1^{-1} = 0.$$

Thus, α is a root of F_n. ■

This parameterization gives a convenient description of the roots of Fibonacci polynomials in terms of roots of unity.

2 Even Dominating Sets

We identify a subset $S \subseteq V$ with its indicator vector $\mathbf{1}_S \in \mathbb{F}_2^V$. Under this identification, symmetric difference of subsets corresponds to addition in $\mathbb{F}_2^V$. The map Φ_G is linear since the state change of each vertex is determined by the parity of the clicks on its closed neighborhood. Consequently, two configurations $V_1, V_2 \subseteq V$ have the same image under Φ_G if and only if $V_1 \triangle V_2 \in \ker \Phi_G$. Thus, configurations producing the same state change form cosets of $\ker \Phi_G$. In particular, whenever an initial configuration is solvable, the set of all solutions is a coset of $\ker \Phi_G$, and hence has 2^d elements, where $d = \dim(\ker \Phi_G)$. $\ker \Phi_G$ also has a natural interpretation in terms of even dominating sets.

Definition 2.1. *An even dominating set of a graph $G = (V, E)$ is a subset $V' \subseteq V$ such that the closed neighborhood $N[v]$ of every vertex contains an even number of vertices from V'.*

Odd dominating sets can be analogously defined. Sutner specifically studied odd dominating sets as the solution to *Lights Out* when all lights are initially on. He showed that such an odd dominating set always exists for any finite graph [Sut89].

Lemma 2.2. *$V' \in \ker \Phi_G$ if and only if V' is an even dominating set of G.*

Proof. Let $V' \in \ker \Phi_G$. Then every vertex is toggled an even number of times when the vertices in V' are clicked. Since each vertex is toggled exactly when the vertices in its closed neighborhood are clicked, for every $v \in V$,

$$|N[v] \cap V'| \text{ is even.}$$

Thus, V' is an even dominating set of G. The converse follows by reversing the same argument. ■

We may now use our graph-theoretic understanding of $\ker \Phi_G$ as even dominating sets to construct even dominating sets on larger grids by tiling those from smaller grids.

Theorem 2.3. *For all $n, k \in \mathbb{N}$,*

$$d(nk - 1) \geq d(n - 1).$$

Proof. Let G_L be an $nk-1 \times nk-1$ grid and G_ℓ be an $n-1 \times n-1$ grid. Since $nk-1 = k(n-1)+(k-1)$, an G may be broken into a $k \times k$ grid of $n-1 \times n-1$ grids, each separated by a horizontal or vertical strip of size 1.

Let Q be an even dominating set of G_ℓ. Let R be Q tiled k times horizontally and vertically, with size 1 separators between each tile, onto G_L where each tile is the reflection across its neighboring separators, as shown below.

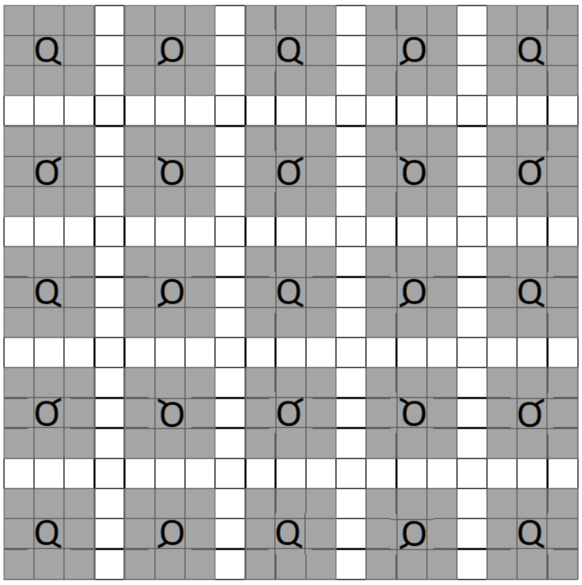

Figure 2: Pattern Q tiled 5 times.

Consider a vertex v in G_L. There are 3 cases.

1. v is within a tile. Then $|N[v] \cap R|$ is even because Q is an even dominating set. Vertices next to tile separators are unaffected by the tiling because the separator vertices are not included in R and thus don't contribute to the parity of $N[v] \cap R$. Thus, $|N[v] \cap R|$ has the same parity as the corresponding closed neighborhood in Q, which is even.

2. v is on a separator between two tiles. Then only two elements in $N[v]$, the vertices within a tile, are also in R. Because neighboring tiles are reflections of each other across separators, these two vertices are either both in R or both not in R. In either case, $|N[v] \cap R|$ is even.

3. v is in the intersection of two separators. Then all of $N[v]$ is in a separator and thus not in R, so $|N[v] \cap R| = 0$, which is even.

Since for every vertex v of G_L, $|N[v] \cap R|$ is even, so R is an even dominating set of G_L. Thus, for every even dominating set of G_ℓ, there exists an even dominating set of G_L obtained through tiling. This tiling function is also injective because we can recover Q by restricting R to just the top left tile. Thus,

$$d(nk-1) \geq d(n-1),$$

as desired. ■

Sutner shows a similar result [Sut88].

The same inequality also follows from the Fibonacci polynomial characterization.

Proof. Since n divides nk, Lemma 1.3 tells us $F_n(x)$ divides $F_{nk}(x)$. Substituting $x+1$, we also have $F_n(x+1)$ divides $F_{nk}(x+1)$. Thus, the first GCD divides the second:

$$\gcd\left(F_n(x), F_n(x+1)\right) \mid \gcd\left(F_{nk}(x), F_{nk}(x+1)\right).$$

So we may write

$$\gcd\left(F_{nk}(x), F_{nk}(x+1)\right) = \gcd\left(F_n(x), F_n(x+1)\right) Q(x),$$

for some $Q(x) \in \mathbb{F}_2[x]$. Taking the degrees, we obtain $d(nk-1) = d(n-1) + \deg Q(x)$, and the desired inequality follows. ■

Note that although the smaller GCD polynomial divides the larger, it's not the case that $d(n-1)$ divides $d(nk-1)$. For example, if $n = 5$ and $k = 6$, then $d(n-1) = d(4) = 4$, which does not divide $d(nk-1) = d(29) = 10$.

3 Number Theory

We now use results from number theory to study the GCDs of Fibonacci polynomials and, consequently, the values of $d(n)$. Many of the results in this section first appeared in the author's earlier works [Boy22a, Boy22b]. We restate several proofs here for completeness and to correct minor errors in those earlier versions.

Lemma 3.1 (Ore)**.** *Let a, b, c, and d be integers. Let (a, b) denote* $\gcd(a, b)$. *Then*

$$(ab, cd) = (a, c)(b, d) \left(\frac{a}{(a,c)}, \frac{d}{(b,d)} \right) \left(\frac{c}{(a,c)}, \frac{b}{(b,d)} \right).$$

We will also use the analogous identity in $\mathbb{F}_2[x]$. Since $\mathbb{F}_2[x]$ is a Euclidean domain, the same proof applies with divisibility and GCDs interpreted in $\mathbb{F}_2[x]$ [O48, Section 5-4, Problem 2, p. 109].

We can now combine the preceding polynomial identity to obtain a recurrence for $d(n)$.

Theorem 3.2. *For all $n \in \mathbb{N}$,*

$$d(2n+1) = 2d(n) + \delta_n, \qquad \delta_n = \begin{cases} 2 & 3 \mid n+1 \\ 0 & \textit{otherwise} \end{cases}.$$

Proof. Let (a, b) denote $\gcd(a, b)$. Applying Lemmas 1.10 and 3.1,

$$\begin{aligned}
d(2n+1) &= \deg\left(F_{2n+2}(x), F_{2n+2}(x+1)\right) \\
&= \deg\left(xF_{n+1}(x)^2, (x+1)F_{n+1}(x+1)^2\right) \\
&= \deg(x, x+1)\left(F_{n+1}(x)^2, F_{n+1}(x+1)^2\right) \\
&\quad \times \left(\frac{x+1}{(x,x+1)}, \frac{F_{n+1}(x)^2}{(F_{n+1}(x)^2, F_{n+1}(x+1)^2)} \right) \left(\frac{x}{(x,x+1)}, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x)^2, F_{n+1}(x+1)^2)} \right) \\
&= \deg\left(F_{n+1}(x), F_{n+1}(x+1)\right)^2 \left(x+1, \frac{F_{n+1}(x)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right) \left(x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right) \\
&= 2d(n) + 2\deg\left(x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right)
\end{aligned}$$

Thus, we have that

$$\delta_n = 2\deg\left(x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right).$$

So, δ_n is 2 exactly when $x+1$ occurs with greater multiplicity in $F_{n+1}(x)$ than in $F_{n+1}(x+1)$. When $n+1$ is not divisible by 3, Corollary 1.4 tells us that $F_{n+1}(x)$ is not divisible by $x+1$, so $\delta_n = 0$. When $n+1$ is divisible by 3, let $n+1 = b2^k$ for $b, k \geq 0$ and b odd. Then b must be divisible by 3. Applying Lemma 1.10,

$$F_{n+1}(x) = x^{2^k-1}F_b(x)^{2^k} \qquad F_{n+1}(x+1) = (x+1)^{2^k-1}F_b(x+1)^{2^k}.$$

Since b is odd, Lemma 1.5 tells us $F_b(x)$ is the square of a square-free polynomial. Since b is a multiple of 3, Corollary 1.4 tells us that $x+1$ divides $F_b(x)$, but $x+1$ does not divide $F_b(x+1)$. So,

$$F_{n+1}(x) = x^{2^k-1}(x+1)^{2^{k+1}}g(x)^{2^{k+1}} \qquad F_{n+1}(x+1) = x^{2^{k+1}}(x+1)^{2^k-1}g(x+1)^{2^{k+1}},$$

for some $g(x) \in \mathbb{F}_2[x]$, where $g(x)$ and $g(x+1)$ are both square-free and divisible by neither x nor $x+1$. So, we see that $F_{n+1}(x)$ can be divided without remainder by $x+1$ more times than $F_{n+1}(x+1)$, so $\delta_n = 2$. ■

This result was first proven by Boyles [Boy22b], resolving a conjecture of Sutner [Sut89].

Repeated application of Theorem 3.2 gives the following explicit formula for $d(n)$.

Corollary 3.3. *Write n as $n+1 = 2^s b$ for $s \geq 0$ and b odd. Then*

$$d(n) = 2^s d(b-1) + \begin{cases} 2(2^s-1) & 3 \mid b \\ 0 & \textit{otherwise} \end{cases}.$$

Proof. We proceed by induction on s. For $s=0$, we have $d(n) = d(b-1)$, which is the desired formula. Now suppose the result holds for some $s \geq 0$. For $n+1 = 2^{s+1}b$, Theorem 3.2 gives

$$\begin{aligned} d(n) &= d\left(2\left(2^s b - 1\right) + 1\right) \\ &= 2d\left(2^s b - 1\right) + \begin{cases} 2 & 3 \mid 2^s b \\ 0 & \text{otherwise} \end{cases}. \end{aligned}$$

Since $3 \mid 2^s b$ if and only if $3 \mid b$, the induction hypothesis gives

$$\begin{aligned} d(n) &= 2\left(2^s d(b-1) + \begin{cases} 2(2^s-1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases}\right) + \begin{cases} 2 & 3 \mid b \\ 0 & \text{otherwise} \end{cases} \\ &= 2^{s+1}d(b-1) + \begin{cases} 2(2^{s+1}-1) & 3 \mid b \\ 0 & \text{otherwise,} \end{cases} \end{aligned}$$

as desired. ■

Thus, the values of $d(n)$ are determined by the values $d(b-1)$ for odd b, together with whether b is divisible by 3.

We can also use Theorem 3.2 to further restrict when $d(n) = 2$.

Corollary 3.4. *If $d(n) = 2$, then $n \equiv -1 \pmod 6$.*

Proof. Since 2 is not divisible by 4, Corollary 1.8 tells us that n must be odd. So, we can write $n = 2^k b - 1$ for b odd and $k \geq 1$. Using Corollary 3.3, we find

$$d(2^k b - 1) = \begin{cases} 2^k d(b-1) + 2(2^k-1) & 3 \mid b \\ 2^k d(b-1) & \text{otherwise} \end{cases}.$$

If $3 \nmid b$, then we must have $d(n) = 2 = 2^k d(b-1)$. This is impossible because Corollary 1.8 tells us that $d(b-1)$ is divisible by 4.

If $3 \mid b$, then we may write $b = 3j$. We also must have $d(n) = 2 = 2^k d(b-1) + 2(2^k-1)$. If $k \geq 2$, then this expression is greater than 2. Thus, we must have $k=1$ and $2 = 2d(b-1)+2$, which simplifies to

$d(b-1)=0$. Solving for n, we find

$$\begin{aligned} n &= 2^k b - 1 \\ &= 2b - 1 \\ &= 6j - 1, \end{aligned}$$

so $n \equiv -1 \pmod 6$, as desired. ■

The preceding results can help us find values of $d(n)$ from smaller known values, but they do not rule out any values of $d(n)$. To obtain such restrictions, we need to understand when individual polynomial factors can occur in the Fibonacci polynomial sequence. We introduce the notation of rank to record the first index at which a given polynomial appears as a factor.

Definition 3.5. *Let P be a non-zero polynomial in $\mathbb{F}_2[x]$. Then*

$$\operatorname{rank}(P) = \min\{r > 0 : P(x) \mid F_r(x)\}.$$

Hunziker, Machiavelo, and Park proved that every polynomial divides some $F_r(x)$ and therefore has a finite rank [HMP04, Corollary 2.8].

We now show a relationship between the divisibility of polynomials and the divisibility of their ranks.

Lemma 3.6. *For any non-zero $P(x) \in \mathbb{F}_2[x]$,*

$$P(x) \mid F_m(x) \iff \operatorname{rank}(P) \mid m.$$

Proof. Let $r = \operatorname{rank}(P)$.

Assume that $P(x)$ divides $F_m(x)$. $P(x)$ must also divide $F_r(x)$. So by Lemma 1.3,

$$\begin{aligned} &P(x) \mid \gcd(F_r(x), F_m(x)) \\ &P(x) \mid F_{\gcd(r,m)}(x). \end{aligned}$$

Since r is minimal, $\gcd(r,m) = r$, and so $r \mid m$.

Now assume that $r \mid m$. Then $\gcd(r,m) = r$. By Lemma 1.3, $F_r(x) \mid F_m(x)$, and so $P(x) \mid F_m(x)$. ■

Thus, the rank of a polynomial completely determines the indices at which it can divide a Fibonacci polynomial. We can now use the ranks to restrict the possible factors of the GCD defining $d(n)$, allowing us to rule out the smallest even value not attained by $d(n)$.

Theorem 3.7. *The smallest even k such that for all $n \in \mathbb{N}$, $d(n) \neq k$ is $k = 26$.*

Proof. Recall from Lemma 1.7 that $d(n)$ is always even. The following computations show that every positive even value less than 26 occurs:

$$\begin{array}{cccccc} d(5)=2 & d(4)=4 & d(11)=6 & d(16)=8 & d(29)=10 & d(84)=12 \\ d(23)=14 & d(19)=16 & d(101)=18 & d(30)=20 & d(59)=22 & d(62)=24. \end{array}$$

We will show that $d(n) = 26$ is impossible.

Assume for the sake of contradiction that there exists $n \in \mathbb{N}$ such that $d(n) = 26$. Since 26 is not divisible by 4, Corollary 1.8 says n must be odd. So, we may write $n = 2m+1$. By Theorem 3.2,

$$26 = d(n) = d(2m+1) = 2d(m) + \begin{cases} 2 & 3 \mid m+1 \\ 0 & \text{otherwise} \end{cases}.$$

Since Lemma 1.7 tells us that $d(m) = 13$ is not possible, we must have $d(m) = 12$ and $m + 1$ is divisible by 3.

Let $m + 1 = 2^s b$ for odd b divisible by 3 and $s \geq 0$. By Corollary 3.3,

$$\begin{aligned} d(m) = 12 &= 2^s d(b-1) + 2(2^s - 1) \\ 14 &= 2^s(d(b-1) + 2) \\ 7 &= 2^{s-1}(d(b-1) + 2). \end{aligned}$$

The only possible integer solutions here are $d(b-1) = 12$, $s = 0$ or $d(b-1) = 5$, $s = 1$. Lemma 1.7 tells us that $d(b-1) = 5$ is impossible, so the only remaining possibility is that $d(m) = d(b-1) = 12$ and $s = 0$. Thus, m must be even.

Since $m + 1$ is odd, Lemma 1.5 tells us that there exists some square-free $R_{m+1}(x)$ such that

$$F_{m+1}(x) = R_{m+1}(x)^2.$$

Define $H_{m+1}(x)$ such that

$$H_{m+1}(x) = \gcd\left(R_{m+1}(x), R_{m+1}(x+1)\right).$$

Since squaring R_{m+1} gives F_{m+1},

$$H_{m+1}(x)^2 = \gcd\left(F_{m+1}(x), F_{m+1}(x+1)\right).$$

Since $R_{m+1}(x)$ is square-free, $H_{m+1}(x)$ is too. Notice too that $H_{m+1}(x)$ is unaffected by translation: $H_{m+1}(x) = H_{m+1}(x+1)$. Since $d(m)$ is 12, $\deg H_{m+1}(x)$ must be 6. Lemma 1.6 tells us that for $y = x^2 + x$, there exists a square-free degree 3 polynomial $h_{m+1}(y) \in \mathbb{F}_2[y]$ such that

$$H_{m+1}(x) = h_{m+1}(y).$$

Since $m + 1$ is odd, we must also require that $h_{m+1}(0) = 1$. There are only three such polynomials in $\mathbb{F}_2[y]$ that satisfy all these requirements:

$$C_1(y) = y^3 + 1 \qquad C_2(y) = y^3 + y + 1 \qquad C_3(y) = y^3 + y^2 + 1.$$

We will handle these case-by-case.

Calculating the rank of our three candidates for H_{m+1} with respect to x,

$$\begin{aligned} \text{rank}(C_1(y)) &= \text{rank}\left((x^2 + x)^3 + 1\right) &&= 85 \\ \text{rank}(C_2(y)) &= \text{rank}\left((x^2 + x)^3 + (x^2 + x) + 1\right) &&= 63 \\ \text{rank}(C_3(y)) &= \text{rank}\left((x^2 + x)^3 + (x^2 + x)^2 + 1\right) &&= 63 \end{aligned}$$

We'll show that none of these are viable candidates for H_{m+1}

Assume for the sake of contradiction that $C_1 = H_{m+1}$. Then $C_1 \mid H_{m+1}$. Since $H_{m+1} \mid R_{m+1} \mid F_{m+1}$, then $C_1 \mid F_{m+1}$. Then by Lemma 3.6, $\text{rank}(C_1) = 85 \mid m + 1$. We also showed earlier that $3 \mid m + 1$. So, $\text{lcm}(85, 3) = 255 \mid m + 1$. Other polynomials with ranks dividing 255 must also divide F_{m+1}. Let

$$P(x) = x^4 + x^3 + 1 \text{ and } Q(x) = x^4 + x^3 + x^2 + x + 1.$$

The ranks of $P(x)$ and $Q(x)$ are 15 and 17 respectively, which are both factors of 255. Thus, by Lemma 3.6, $P(x) \mid F_{m+1}(x + 1)$. Substituting $x + 1$ in place of x, we also obtain that $P(x + 1) \mid F_{m+1}(x + 1)$. Since $P(x + 1) = Q(x)$, then $Q(x) \mid F_{m+1}(x)$. Substituting $x + 1$ in place of x, we also obtain that $Q(x + 1) \mid F_{m+1}(x + 1)$. Since P and Q divide both $F_{m+1}(x)$ and $F_{m+1}(x + 1)$, they both also divide the GCD, so P and Q divide H_{m+1}^2. Since P and Q are both square-free, P and Q divide H_{m+1}. Since C_1, P, and Q are all pairwise coprime,

$$\begin{aligned} C_1 P Q &\mid H_{m+1} \\ \deg H_{m+1} &\geq \deg C_1 + \deg P + \deg Q = 14, \end{aligned}$$

which contradicts our assumption that the degree was 6. Thus C_1 is excluded as a candidate.

Assume for the sake of contradiction that $C_2 = H_{m+1}$. Since $C_2 = H_{m+1}$, $C_2 \mid H_{m+1}$. Notice that H_{m+1} must also divide $R_{m+1}(x)$ and $R_{m+1}(x+1)$. Similarly, H_{m+1} divides both $F_{m+1}(x)$ and $F_{m+1}(x+1)$. Thus, $C_2 \mid F_{m+1}(x)$ and $F_{m+1}(x+1)$. By Lemma 3.6, $\operatorname{rank}(C_2) = 63 \mid m+1$. Since C_3 also has rank 63, Lemma 3.6 tells us that $C_3 \mid F_{m+1}(x)$. Since C_3 is translation invariant, substituting $x+1$ for x shows that C_3 also divides $F_{m+1}(x+1)$. Since C_3 divides both $F_{m+1}(x)$ and $F_{m+1}(x+1)$, then C_3 must also divide their GCD H_{m+1}^2. Then since C_3 is square-free, $C_3 \mid H_{m+1}$. Since C_2 and C_3 are coprime,

$$\begin{aligned} C_2C_3 &\mid H_{m+1} \\ \deg H_{m+1} &\geq \deg C_2 + \deg C_3 = 12. \end{aligned}$$

which contradicts our assumption that the degree was 6. Thus C_2 is excluded as a candidate. We can apply the same argument with C_2 and C_3 swapped to show that C_3 may also be excluded as a candidate.

We have shown that no such polynomial H_{m+1} can exist. Therefore, there is no $3 \mid m+1$ with $d(m) = 12$ and hence no n with $d(n) = 26$. ■

Theorem 3.7 gives one impossible value of $d(n)$. The recurrence in Theorem 3.3 allows us to propagate this obstruction to infinitely many more values.

Corollary 3.8. *Suppose $a \in \mathbb{N}$ is even and an impossible value of $d(n)$. Then $2a+2$ is also an impossible value of $d(n)$.*

Proof. Suppose for the sake of contradiction that there exists n such that $d(n) = 2a+2$. Since $2a+2$ is not divisible by 4, then Corollary 1.8 tells us that n must be odd, so we may write $n = 2m+1$. So, by Theorem 3.2,

$$2a+2 = d(n) = d(2m+1) = 2d(m) + \delta_m \qquad \delta_m = \begin{cases} 2 & 3 \mid m+1 \\ 0 & \text{otherwise} \end{cases}.$$

If $\delta_m = 0$, then $2a+2 = 2d(m)$. By Lemma 1.7, $d(m)$ is even, so $2d(m)$ is divisible by 4. This contradicts the fact that $2a+2$ is not divisible by 4 because a is even. If $\delta_m = 2$, then $d(m) = a$, contradicting the impossibility of a as an output of d. ■

Expanding the recurrence, if a is an impossible even value, then $2^t(a+2) - 2$ is also impossible for all $t \geq 0$. In other words, $26, 54, 110, 222, 446, \ldots$ are all impossible values of $d(n)$.

4 Finite Fields

We now turn to finite field methods to study the Fibonacci polynomial GCDs that determine $d(n)$. In particular, we use properties of roots of unity and traces to analyze $d(p^k - 1)$ for prime p and to characterize when $d(p-1) = 0$.

4.1 Orders of Prime Powers

Lemma 4.1. *Let p be an odd prime. Define $h = \operatorname{ord}_p(2)$ and $c = v_p(2^h - 1)$, where $v_p(a)$ is the p-adic valuation of a. Then for every $j \geq 1$,*

$$\operatorname{ord}_{p^j}(2) = hp^{\max(0, j-c)}.$$

Proof. Suppose $2^r \equiv 1 \pmod{p^j}$. Since $2^r \equiv 1 \pmod{p}$, the definition of h gives $h \mid r$, so we may write $r = ht$. By the Lifting the Exponent Lemma [IR90, Chapter 2 Section 2.7],

$$\begin{aligned} v_p(2^r - 1) &= v_p(2^{ht} - 1) \\ &= v_p(2^h - 1) + v_p(t) \\ &= c + v_p(t). \end{aligned}$$

Thus,

$$p^j \mid 2^{ht} - 1 \iff c + v_p(t) \geq j.$$

The smallest possible integer t satisfying this inequality is $t = p^{\max(0, j-c)}$, so

$$\operatorname{ord}_{p^j}(2) = hp^{\max(0, j-c)},$$

as desired. ■

Definition 4.2. *A prime number p is a Wieferich prime when*

$$p^2 \mid 2^{p-1} - 1.$$

For an odd prime p, the Wieferich primes are precisely the primes for which $v_p(2^{\operatorname{ord}_p(2)} - 1) > 1$. Thus, they are exactly the exceptional primes for which the order formulation in Lemma 4.1 does not take its simplest form.

It is conjectured that infinitely many Wieferich primes exist, but only two are currently known: 1093 and 3511 [CDP97]. Computer searches have shown any more Wieferich primes must be greater than 2^{64} [Pri26].

The following equivalent characterization of Wieferich primes will be useful because it expresses the Wieferich condition in terms of the multiplicative order appearing in Lemma 4.1.

Lemma 4.3. *An odd prime number p is a Wieferich prime if and only if*

$$p^2 \mid 2^{\operatorname{ord}_p(2)} - 1.$$

Proof. Let $n = \operatorname{ord}_p(2)$. Then $p - 1 = kn$ for some natural number k, and

$$2^{p-1} - 1 = 2^{kn} - 1.$$

So, by the Lifting the Exponent lemma [IR90, Chapter 2 Section 2.7],

$$v_p(2^{kn} - 1) = v_p(2^n - 1) + v_p(k).$$

Since $1 \leq k \leq p - 1$, we have $v_p(k) = 0$. Thus,

$$v_p(2^{p-1} - 1) = v_p(2^n - 1).$$

So, $p^2 \mid 2^{p-1} - 1$ if and only if $p^2 \mid 2^{\operatorname{ord}_p(2)} - 1$, as desired. ■

Consequently, if p is not a Wieferich prime, then $v_p(2^{\operatorname{ord}_p(2)} - 1) = 1$. So, $c = 1$ and $\operatorname{ord}_{p^j}(2) = \operatorname{ord}_p(2)\, p^{j-1}$.

Corollary 4.4. *Let p be an odd prime. If p is not a Wieferich prime, then*

$$\operatorname{ord}_{p^j}(2) = \operatorname{ord}_p(2)\, p^{j-1}.$$

Proof. By Lemma 4.1,

$$\operatorname{ord}_{p^j}(2) = \operatorname{ord}_p(2)\, p^{\max(0, j-c)}, \qquad c = v_p(2^{\operatorname{ord}_p(2)} - 1).$$

Thus it is sufficient to show that $c = 1$. By definition of $\operatorname{ord}_p(2)$, we have $c \geq 1$. Since p is not a Wieferich prime, Lemma 4.3 implies that

$$p^2 \nmid 2^{\operatorname{ord}_p(2)} - 1.$$

Thus $c < 2$ and therefore must be exactly 1, as desired. ■

4.2 Extension Degrees & Traces

We next turn to finite field extensions and traces. These tools will allow us to relate the roots of Fibonacci polynomials to their behavior over finite fields.

Definition 4.5. *Let b be coprime to n. Then signed order of b modulo n is*

$$\rho_b(n) = \min\{r > 0 : b^r \equiv \pm 1 \pmod{n}\}.$$

This is also sometimes called the *index* of " ± 1 relative to b, written $\operatorname{ind}_b(\pm 1)$. Note that compared to the typical order, the signed order is either the same value or half as much.

Lemma 4.6. *Let p be an odd prime, $j \geq 1$, and $H_j = \operatorname{ord}_{p^j}(2)$. Then*

$$\rho_2(p^j) = \begin{cases} H_j/2 & H_j \text{ is even} \\ H_j & H_j \text{ is odd} \end{cases}.$$

Moreover, when H_j is even,

$$2^{H_j/2} \equiv -1 \pmod{p^j}.$$

Proof. Assume H_j is even. Then by definition of H_j,

$$\left(2^{H_j/2}\right)^2 \equiv 1 \pmod{p^j}.$$

Thus the only solutions are $2^{H_j/2} \equiv \pm 1 \pmod{p^j}$. However, 1 cannot be a solution; otherwise, H_j would not be minimal. So,

$$2^{H_j/2} \equiv -1 \pmod{p^j}.$$

Notice that if $r \leq H_j/2$, and $2^r \equiv -1 \pmod{p^j}$, then $2^{2r} \equiv 1 \pmod{p^j}$; since H_j is minimal, $H_j \leq 2r$. Thus to satisfy both constraints we must have $H_j/2 = r$.

Now assume H_j is odd. Notice that by definition,

$$H_j = \operatorname{ord}_{p^j}(2) = \min\{h > 0 : 2^h \equiv 1 \pmod{p^j}\},$$

so smaller solutions can only be of the form $2^r \equiv -1 \pmod{p^j}$. Then $2^{2r} \equiv 1 \pmod{p^j}$, which means $H_j \mid 2r$, and since H_j is odd, $H_j \mid r$. Thus $H_j \leq r$, so H_j is minimal, as desired. ∎

We can use the signed order to find the degree of certain extensions of $\mathbb{F}_2$.

Lemma 4.7. *Let $M > 1$ be odd; let ζ have exact order M; set $\alpha = \zeta + \zeta^{-1}$. Then*

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M).$$

Proof. A standard finite field fact says

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \min\{r > 0 : \alpha^{2^r} = \alpha\}$$

[LN97, Theorem 2.14, p. 52]. By definition of α, $\alpha^{2^r} = \zeta^{2^r} + \zeta^{-2^r}$, so $\alpha^{2^r} = \alpha$ is equivalent to $\zeta^{2^r} + \zeta^{-2^r} = \zeta + \zeta^{-1}$.

For any non-zero $a, b \in \overline{\mathbb{F}_2}$, $a + a^{-1} = b + b^{-1} \iff (a+b)(ab+1) = 0$. So, either $a = b$ or $a = b^{-1}$. Taking $a = \zeta^{2^r}$ and $b = \zeta$, we obtain

$$\zeta^{2^r} = \zeta \quad \text{or} \quad \zeta^{2^r} = \zeta^{-1}.$$

Since ζ has exact order M, we can equivalently write

$$2^r \equiv 1 \pmod{M} \quad \text{or} \quad 2^r \equiv -1 \pmod{M}.$$

So, $\alpha^{2^r} = \alpha \iff 2^r \equiv \pm 1 \pmod{M}$, and

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \min\{r > 0 : \alpha^{2^r} = \alpha\} = \rho_2(M),$$

as desired. ∎

We will also need the trace of elements in finite field extensions. We first recall its standard expression in terms of the Frobenius automorphism [LN97, Definition 2.22, p. 54].

Definition 4.8 (Lidl & Niederreiter)**.** *Let a be algebraic over $\mathbb{F}_q$. Then for every $z \in \mathbb{F}_q(a)$,*

$$\mathrm{Tr}_{\mathbb{F}_q(a)/\mathbb{F}_q}(z) = \sum_{r=0}^{[\mathbb{F}_q(a):\mathbb{F}_q]-1} z^{q^r}.$$

Corollary 4.9.

$$\mathrm{Tr}_{\mathbb{F}_q(a)/\mathbb{F}_q}(1) = [\mathbb{F}_q(a) : \mathbb{F}_q].$$

Proof. Immediate from evaluating Definition 4.8 at $z = 1$. ∎

Lemma 4.10. *Let $M > 1$ be odd, and let ζ have exact order M. Let $\alpha = \zeta + \zeta^{-1}$. Define*

$$G_M = \{\pm 2^r \pmod{M} : r \geq 0\} \subseteq (\mathbb{Z}/M\mathbb{Z})^\times.$$

Then

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u.$$

Proof. Lemma 4.7 showed that

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M).$$

So, by Definition 4.8,

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{r=0}^{\rho_2(M)-1} \zeta^{2^r} + \zeta^{-2^r}.$$

By the definition of $\rho_2(M)$, the residues $2^0, -2^0, 2^1, -2^1, \ldots, 2^{\rho_2(M)-1}, -2^{\rho_2(M)-1}$. are all distinct modulo M. Otherwise, we would obtain $2^r \equiv \pm 2^s \pmod{M}$, or equivalently, $2^{r-s} \equiv \pm 1 \pmod{M}$ for some $0 \leq s < r < \rho_2(M)$, which contradicts $\rho_2(M)$ being minimal.

The residues are exactly the elements of G_M, so

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u,$$

as desired. ∎

Corollary 4.11. *If $G_M = (\mathbb{Z}/M\mathbb{Z})^\times$, then*

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \mu(M) \pmod{2},$$

where μ is the number-theoretic Möbius function.

Proof. From Lemma 4.10, we know that

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u.$$

When $G_M = (\mathbb{Z}/M\mathbb{Z})^\times$, we also have

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{0 \le u < M, \gcd(u,M)=1} \zeta^u = \mu(M) \pmod 2,$$

by Ramanujan's sum identity of roots of unity, as desired. ■

4.3 Applications to $d(n)$

We can now combine our previous results to greatly reduce the calculation of $d(p^k - 1)$ when p is a prime number.

Theorem 4.12. *Let p be an odd prime and set $c = v_p(2^{\mathrm{ord}_p(2)} - 1)$. Then for every $k \ge c$,*

$$d(p^k - 1) = d(p^c - 1).$$

Proof. Fix $k \ge c$. For $1 \le j \le k$, let

$$R_j = \{\zeta + \zeta^{-1} : \mathrm{ord}(\zeta) = p^j\}.$$

By Lemma 1.11,

$$\mathrm{roots}\, F_{p^k} = \bigcup_{j=1}^{k} R_j.$$

Suppose $\alpha, \alpha + 1 \in \overline{\mathbb{F}_2}$ are both roots of F_{p^k}, with

$$\alpha \in R_j, \qquad \alpha + 1 \in R_\ell.$$

Because $\mathbb{F}_2(\alpha) = \mathbb{F}_2(\alpha+1)$, their extension degrees over $\mathbb{F}_2$ are equal. By Lemma 4.7,

$$\begin{aligned}[]
[\mathbb{F}_2(\alpha) : \mathbb{F}_2] &= [\mathbb{F}_2(\alpha+1) : \mathbb{F}_2] \\
\rho_2(p^j) &= \rho_2(p^\ell) \\
\rho_2(p)p^{\max(0, j-c)} &= \rho_2(p)p^{\max(0,\ell-c)}.
\end{aligned}$$

Hence either $j, l \le c$ or $j = \ell > c$.

It remains to exclude the second possibility. Assume for the sake of contradiction that $j = \ell > c$. Then

$$\alpha, \alpha + 1 \in R_\ell = R_j.$$

Set

$$M = p^j, \qquad m = p^{j-c},$$

and choose a primitive Mth root $\zeta \in \overline{\mathbb{F}_2}$ such that

$$\alpha = \zeta + \zeta^{-1}.$$

Since $\alpha + 1 \in R_j$, there must exist an integer s with $\gcd(s, p) = 1$ such that

$$\alpha + 1 = \zeta^s + \zeta^{-s}$$

Thus

$$1 + \alpha + \alpha + 1 = 0$$
$$1 + \zeta + \zeta^{-1} + \zeta^s + \zeta^{-s} = 0,$$

so ζ is a root of

$$P_s(x) = 1 + x + x^{M-1} + x^s + x^{M-s}. \tag{1}$$

The element ζ^m has exact order p^c. Let $f(x)$ be the minimal polynomial of ζ^m over $\mathbb{F}_2$. Then

$$\deg f = \mathrm{ord}_{p^c}(2) = \mathrm{ord}_p(2)\,.$$

Since ζ^m is a root of $f(x)$, ζ is a root of $f(x^m)$, and

$$\deg f(x^m) = \mathrm{ord}_p(2)\, m = \mathrm{ord}_p(2)\, p^{j-c} = \mathrm{ord}_{p^j}(2)\,.$$

This is exactly the degree of the minimal polynomial of ζ, so $f(x^m)$ is that minimal polynomial, and

$$f(x^m) \mid P_s(x).$$

Every polynomial has a unique decomposition by exponents modulo m:

$$P_s(x) = \sum_{a=0}^{m-1} x^a Q_a(x^m).$$

If $f(x^m) \mid P_s(x)$, uniqueness implies that $f(x^m)$ divides every $Q_a(x^m)$. However, only the constant exponent in (1) is divisible by m. The other four exponents are congruent modulo m to ± 1 and $\pm s$ are are non-zero because $p \nmid s$. Therefore $Q_0(x^m) = 1$, which is constant. However, $f(x^m)$ is not constant and thus cannot divide $Q_0(x^m)$, a contradiction.

Such a contradiction excludes every common root in R_j for $j > c$. All common roots of $F_{p^k}(x)$ and $F_{p^k}(x+1)$ therefore lie in

$$\bigcup_{j=1}^{c} R_j = \mathrm{roots}\, F_{p^c}.$$

Theorem 2.3 also tells us that such common roots are also common roots for larger powers of p. Lemma 1.7 tells us that every root has multiplicity 2, so their GCDs have the same degree. Thus,

$$d(p^k - 1) = d(p^c - 1),$$

as desired. ∎

The result simplifies even further when p is not a Wieferich prime.

Corollary 4.13. *If p is a non-Wieferich prime, then for every $k \geq 1$,*

$$d(p^k - 1) = d(p - 1).$$

Proof. The case for $p = 2$ is the known identity $d(2^k - 1) = 0$ [Sut89]. For an odd non-Wieferich prime,

$$c = v_p\,(2 - 1) = 1,$$

so the desired result follows directly from Theorem 4.12. ∎

We can also use Theorem 4.12 to show the result holds for the two known non-Wieferich primes.

Corollary 4.14. *For all $k \geq 1$,*

$$d(1093^k - 1) = d(3511^k - 1) = 0.$$

Proof. Calculating the relevant stabilization exponent for each prime,

$$c_{1093} = v_{1093}\left(2^{\mathrm{ord}_{1093}(2)} - 1\right) = v_{1093}\left(2^{364} - 1\right) = 2$$
$$c_{3511} = v_{3511}\left(2^{\mathrm{ord}_{3511}(2)} - 1\right) = v_{3511}\left(2^{1755} - 1\right) = 2.$$

Applying Theorem 4.12 and computing powers 1 and 2,

$$d(1093^1 - 1) = d(1093^2 - 1) = 0$$
$$d(3511^1 - 1) = d(3511^2 - 1) = 0,$$

and the desired result follows. ■

Thus the result holds the same for every known prime, leading us the the following conjecture.

Conjecture 4.15. *For all primes p and all $k \geq 0$,*

$$d(p^k - 1) = d(p - 1).$$

From Corollary 4.15 and the currently known Wieferich primes, any counter-example would require a newly discovered Wieferich prime greater than 2^{64} [Pri26].

We can combine this result with our recurrence for $d(2n+1)$ to obtain an explicit formula for a broad family of indices.

Corollary 4.16. *Let p be an odd non-Wieferich prime. Then for all $n, m \geq 0$,*

$$d(2^n p^m - 1) = \begin{cases} 0 & m = 0 \\ 2^n d(p-1) & m \geq 1, p \neq 3 \\ 2^{n+1} - 2 & m \geq 1, p = 3 \end{cases}.$$

Proof. The $m = 0$ case reduces to what Sutner already showed [Sut89]. For the other cases, we first apply Corollary 3.3 to obtain

$$d(2^n p^m - 1) = \begin{cases} 0 & m = 0 \\ 2^n d(p^m-1) & m \geq 1, p \neq 3 \\ 2^{n+1} - 2 & m \geq 1, p = 3 \end{cases}.$$

We then apply Corollary 4.15 to change $d(p^m - 1)$ to $d(p-1)$. ■

Corollary 4.17. *For all $n, m \geq 1$, the following relationships hold*

$$\begin{array}{ccc}
d(3^n - 1) = 0 & d(5^n - 1) = 4 & d(7^n - 1) = 0 \\
d(11^n - 1) = 0 & d(13^n - 1) = 0 & d(17^n - 1) = 8 \\
d(19^n - 1) = 0 & d(23^n - 1) = 0 & d(29^n - 1) = 0 \\
d(31^n - 1) = 20 & d(37^n - 1) = 0 & d(41^n - 1) = 0 \\
d(43^n - 1) = 0 & d(47^n - 1) = 0 & d(53^n - 1) = 0 \\
d(59^n - 1) = 0 & d(61^n - 1) = 0 & d(67^n - 1) = 0 \\
d(71^n - 1) = 0 & d(73^n - 1) = 0 & d(79^n - 1) = 0 \\
d(83^n - 1) = 0 & d(89^n - 1) = 0 & d(97^n - 1) = 0 \\
d(127^n - 1) = 56 & d(257^n - 1) = 144 & d(683^n - 1) = 132 \\
d(2731^n - 1) = 468 & d(8191^n - 1) = 4004 & d(43691^n - 1) = 6936,
\end{array}$$

and

$$\begin{array}{lll}
d(2^n 3^m - 1) = 2^{n+1} - 2 & d(2^n 5^m - 1) = 2^{n+2} & d(2^n 7^m - 1) = 0 \\
d(2^n 11^m - 1) = 0 & d(2^n 13^m - 1) = 0 & d(2^n 17^m - 1) = 2^{n+3} \\
d(2^n 19^m - 1) = 0 & d(2^n 23^m - 1) = 0 & d(2^n 29^m - 1) = 0 \\
d(2^n 31^m - 1) = 2^{n+2} \cdot 5 & d(2^n 37^m - 1) = 0 & d(2^n 41^m - 1) = 0 \\
d(2^n 43^m - 1) = 0 & d(2^n 47^m - 1) = 0 & d(2^n 53^m - 1) = 0 \\
d(2^n 59^m - 1) = 0 & d(2^n 61^m - 1) = 0 & d(2^n 67^m - 1) = 0 \\
d(2^n 71^m - 1) = 0 & d(2^n 73^m - 1) = 0 & d(2^n 79^m - 1) = 0 \\
d(2^n 83^m - 1) = 0 & d(2^n 89^m - 1) = 0 & d(2^n 97^m - 1) = 0 \\
d(2^n 127^m - 1) = 2^{n+3} \cdot 7 & d(2^n 257^m - 1) = 2^{n+4} \cdot 9 & d(2^n 683^m - 1) = 2^{n+2} \cdot 33 \\
d(2^n 2731^m - 1) = 2^{n+2} \cdot 117 & d(2^n 8191^m - 1) = 2^{n+2} \cdot 1001 & d(2^n 43691^m - 1) = 2^{n+3} \cdot 867.
\end{array}$$

Per OEIS sequence A159257, the above lists all odd primes less than 100 explicitly and all primes less than 25000 for which $d(p-1) \neq 0$. We calculated $d(43691-1)$ ourselves via computer program.

We can also use the trace results to identify primes for which $d(p-1) = 0$.

Theorem 4.18. *Let $p \equiv 3 \pmod 4$ be a prime where*

$$\{\pm 2^r \pmod p : r \geq 0\} = (\mathbb{Z}/p\mathbb{Z})^\times,$$

or equivalently,

$$\mathrm{ord}_p(2) \in \left\{p-1, \frac{p-1}{2}\right\}$$

Then $d(p-1) = 0$.

Proof. Assume for the sake of contradiction that $d(p-1) \neq 0$. Then $F_p(x)$ and $F_p(x+1)$ have a common factor and thus a common root α, so

$$F_p(\alpha) = F_p(\alpha+1) = 0.$$

By Lemma 1.11, there must exist non-trivial pth roots of unity ζ and η such that

$$\alpha = \zeta + \zeta^{-1} \qquad \alpha + 1 = \eta + \eta^{-1}.$$

We can see that $\mathbb{F}_2(\alpha+1) = \mathbb{F}_2(\alpha)$ (α and $\alpha+1$ can be obtained from each other by adding 1), and so by Corollary 4.11,

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha+1)/\mathbb{F}_2}(\alpha+1) = \mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \mu(p) \pmod 2 = 1.$$

However, by Lemmas 4.7 and 4.6,

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = [\mathbb{F}_2(\alpha+1) : \mathbb{F}_2] = \rho_2(p) = \frac{p-1}{2}.$$

Since $p \equiv 3 \pmod 4$, this value is odd. Applying the linearity of the trace and Corollary 4.9,

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha+1) = \mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) + \mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(1) = \mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) + 1 = 0.$$

Thus we have reached a contradiction. ■

Taken together, these results give several ways to determine and constrain the values of $d(n)$. In particular, the behavior of $d(n)$ at prime powers is controlled by the arithmetic of 2 modulo those primes, with Wieferich primes providing the principal exceptional case. The final theorem also gives a broad family of primes for which $d(p-1) = 0$.

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